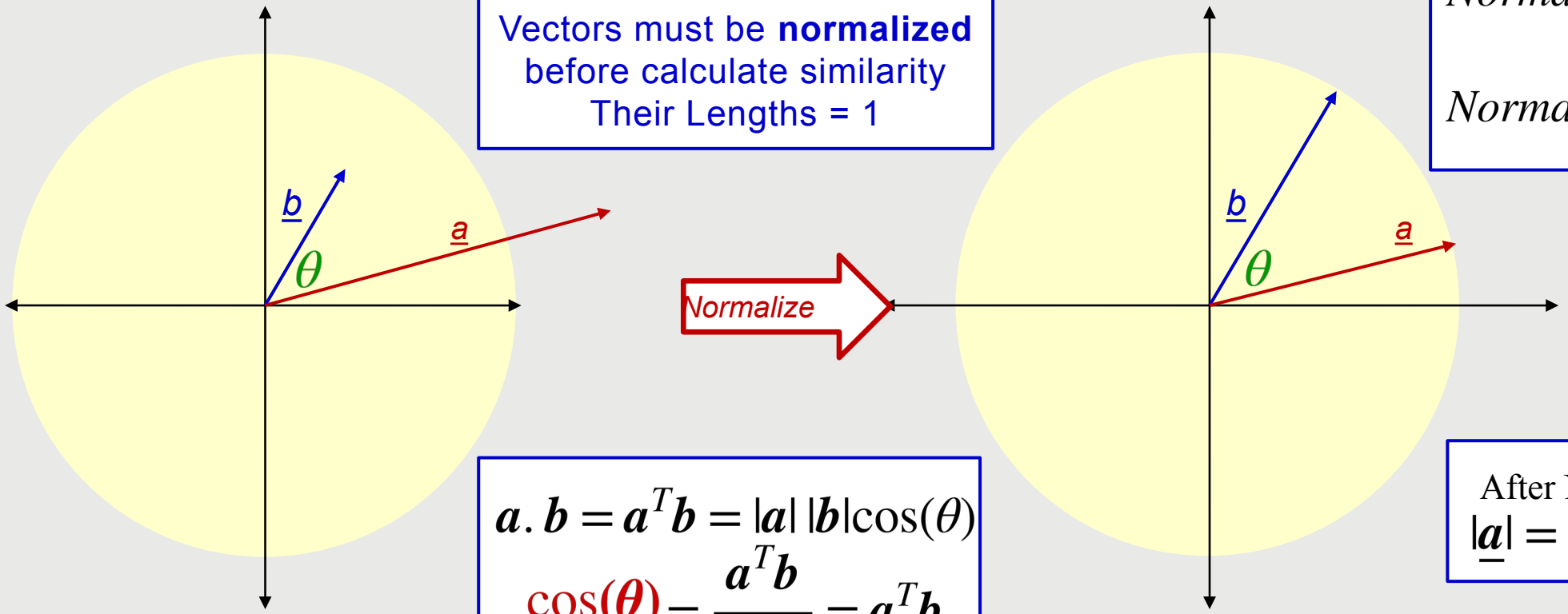
Review: Cosine Similarity

Vectors must be **normalized** before calculate similarity
Their Lengths = 1

$$Normalized\ \underline{a} = \frac{\underline{a}}{|\underline{a}|}$$

$$Normalized\ \underline{b} = \frac{\underline{b}}{|\underline{b}|}$$



$$\underline{a}.\underline{b} = \underline{a}^T\underline{b} = |\underline{a}|\,|\underline{b}|\cos(\theta)$$

$$\cos(\theta) = \frac{\underline{a}^T\underline{b}}{|\underline{a}|\,|\underline{b}|} = \underline{a}^T\underline{b}$$

After Normalization
$|\underline{a}| = |\underline{b}| = 1$

Cosine Similarity is between **-1 , +1**
It is a dot product between two Normalized vectors **a** and **b**

Maximize [Similarity] = Minimize -[Similarity]

Loss = -[Similarity]